SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

EAS109

Examination - July Semester 2025

Thermo-Fluid Mechanics

Friday, 14 November 2025 **10:00 am - 12:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **SIX (6)** printed pages (including cover page).
2. You must answer **ALL** questions.
3. All answers must be written in the answer book.
4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Answer all questions. (Full marks: 100)

Question 1

Oil of relative density 0.87 is siphoned from a tank as shown in Figure Q1. The maximum siphon height is 3.5 m above the oil level in the tank. The oil is released to the atmosphere through the pipe outlet which is 10.0 m below the oil level. The pipe has a diameter of 5 cm while the nozzle outlet diameter is 20 mm as shown. All pressure measurements are in gauge pressure.

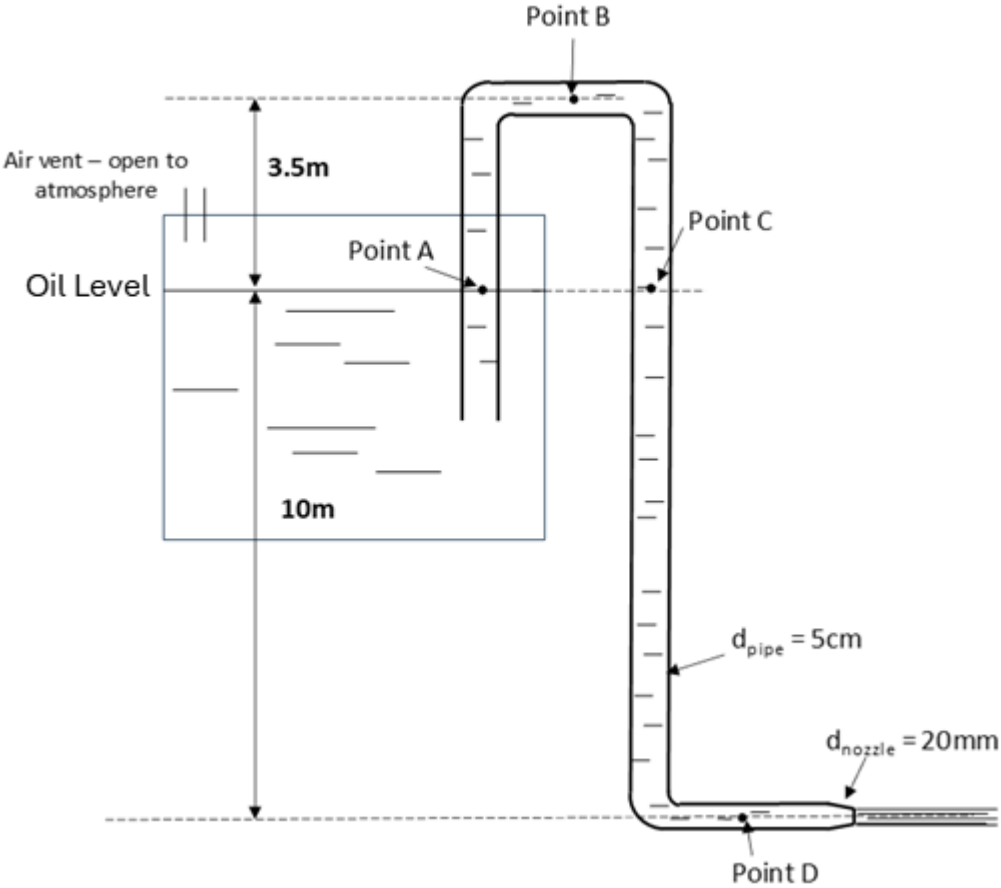

Figure Q1

Neglecting the pipe losses and identifying the basic thermodynamic and fluid equations for the fundamental flow study of this siphon.

Question 1a

Recognise the volumetric flowrate (m^3/s) and the mass flowrate (kg/s) of oil from the tank.

(5 marks)

Question 1b

Interpret the velocity of oil in the pipe in m/s.

(4 marks)

Question 1c

Compute the pressures at the points A, B, C, and D in kPa.

(8 marks)

Question 1d

Describe **FOUR (4)** limitations in application of the Bernoulli Equation.

(8 marks)

Question 2

During a bench test, a basic cycle gas turbine unit operating on the Joule-Brayton cycle draws in air at absolute pressure of 1.0 bar and at a temperature of 25°C. The air is compressed through a compressor unit to a pressure ratio of 9.5:1. It then goes through the combustion chamber where heat of 450 kJ/kg is supplied.

[NOTE: This is an Air Standard Cycle. For air, take γ = 1.4 and Cp = 1.005 kJ/kgK]

Question 2a

Sketch the P-v and T-s diagram of this cycle.

(4 marks)

Question 2b

Interpret the numerical values of the following quantities stated in Question 2b(i) to Question 2b(iii) below:

Question 2b(i)

Temperature (K), at the end of each process for this cycle.

(5 marks)

Question 2b(ii)

Net work done in kJ/kg.

(5 marks)

Question 2b(iii)

Thermal efficiency of the cycle.

(3 marks)

Question 2c

Identify **FOUR (4)** air-standard assumptions when analysing the gas turbine unit above.

(8 marks)

Question 3a

The Hagen-Poiseuille equation for laminar flow is given as $\Delta P = \frac{32\mu L v}{d^2}$. Apply the Hagen-Poiseuille equation in terms of Friction Head Loss, H_f .

By comparing the Hagen-Poiseulle and Darcy's equation, derive the friction factor (f) for laminar flow and recognise that it is $\frac{16}{Re}$.

(10 marks)

Question 3b

Piping details of a refinery facility are shown in Figure Q3b of Appendix 1. In the facility, oil of density 880 kg/m^3 is transferred from Tank A to Tank B through the main pipe and diverged in parallel to Branch 1 and Branch 2 with identical length of 40 m. Point A and Point B are located at an elevation difference of 15 m and have the same volumetric flowrate produced by the pump. The pressure at points A and B are 780 kPa and 650 kPa, respectively. All pressure measurements are in gauge pressure.

Apply Bernoulli Equation, $\frac{P_A}{\rho g} + \frac{v_A^2}{2g} + Z_A - h_{Losses} = \frac{P_B}{\rho g} + \frac{v_B^2}{2g} + Z_B$, (where h_{Losses} is the pipe losses due to friction).

Identify the parameters in Question 3b(i) to Question 3b(iii) using suitable calculations.

Question 3b(i)

The velocity (m/s) of oil in Branch 1 pipe.

(5 marks)

Question 3b(ii)

The velocity (m/s) of oil in Branch 2 pipe.

(5 marks)

Question 3b(iii)

The volumetric flowrate produced by the pump in m^3/hr.

(5 marks)

Question 4

A truck carrying perishable products has a cylindrical refrigerant tank with 2 m length and 0.5 m diameter installed on top of its storage container. Due to the additional headwind loading on the cylinder, the truck engine needs to generate additional power requirement. To minimise fuel consumption, the operator needs to decide the cylinder orientation on the storage container. Assume that the truck is traveling at a maximum speed of 90 km/h on the expressway with ideal atmospheric air conditions of 101.3 kPa and 25°C.

Question 4a

Compute the additional power requirement when the cylinder is installed such that its circular surfaces face:

Question 4a(i)

The front and back of the truck with drag coefficient $C_D = 0.9$.

(8 marks)

Question 4a(ii)

The sides of the truck with drag coefficient $C_D = 0.8$.

(6 marks)

Question 4b

From the solutions for Question 4a(i) and Question 4a(ii), explain which orientation is optimal to minimise fuel consumption.

(5 marks)

Question 4c

Describe **THREE (3)** main assumptions used in your calculations.

(6 marks)

Appendix 1

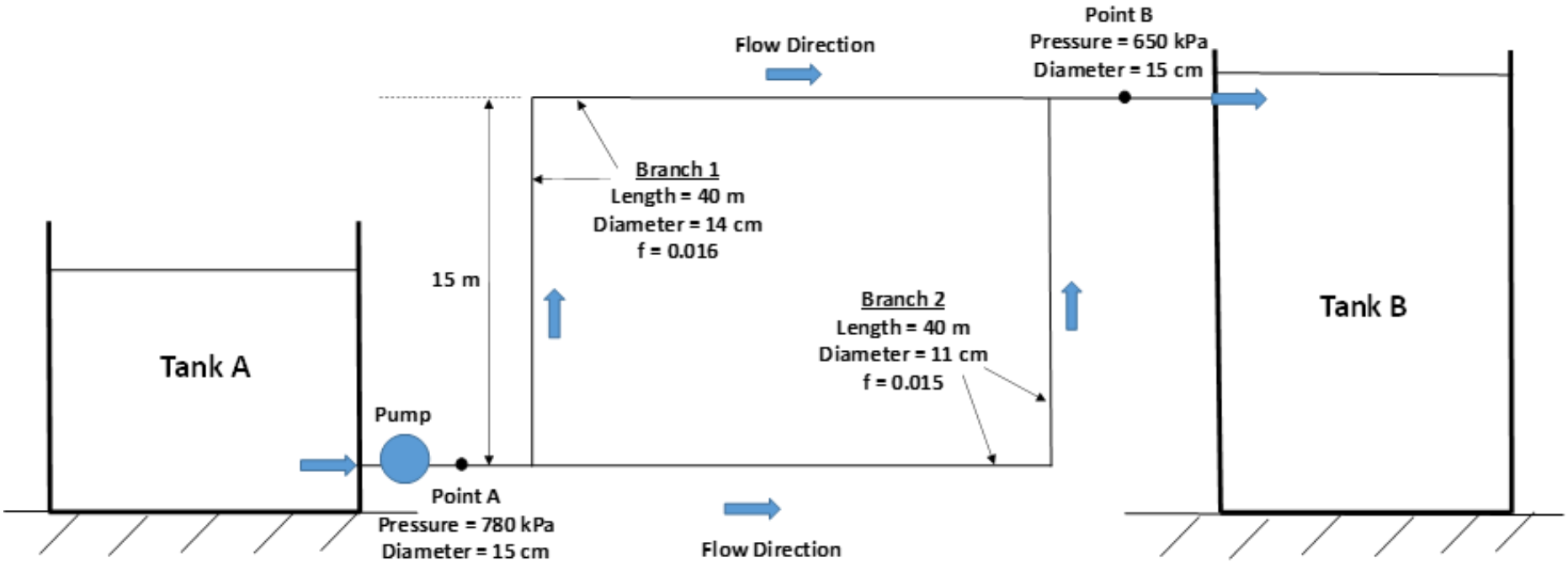

Figure Q3b

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